Student Name: _____

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Deep Learning Midterm (CS 7150)
Fall 2022 **ANSWER KEY**

Answer key. Solutions appear in boxes after each question.

Instructions

- Important: WRITE YOUR NAME and your Northeastern email address on EVERY page.
- Write your answers in the spaces directly after each question.
- You have 90 minutes to do the exam.
- For worked questions, draw a single BOX around your final answer, so that graders can tell the difference between your answer and your scratch work.
- You may use the supplied scratch paper. Let the proctor know if you need more. Scratch calculations will not be graded.
- All cell phones and other electronic devices must be turned off during this exam.
- No outside aids or resources are allowed. You are not allowed to use your notes, books, computers, calculators, or any other resource.
- You are not allowed to communicate with any person other than the instructor / exam proctor while taking this exam.
- You may not share, disseminate, or discuss these questions with any other student in this course who has not taken the exam yet. Doing so is considered academic dishonesty and will lead to nullification of exam grades.
- This midterm exam may not be posted online, copied, or disseminated.

Signature

Sign below to attest that you have read and will follow the closed-book exam rules above.

Student signature: _____ Date: _____

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1. True or false, circle one.

   (a) A two-layer neural network can exactly match any binary logic truth table, given enough neurons. True / False.

   > **Solution: True**: write the table in disjunctive normal form: one hidden threshold unit per true row (an AND of the literals), and an output unit that ORs them.

   (b) There are continuous functions that no two-layer neural network can approximate closely regardless of how many neurons are used. True / False.

   > **Solution: False**: by universal approximation (Cybenko), a two-layer network with enough hidden units can approximate any continuous function on a compact domain as closely as we like.

   (c) A deep linear network wihout nonlinearities can compute the same functions as a network with nonliearities, if given enough neurons. True / False.

   > **Solution: False**: a composition of linear maps is a single linear map, so depth adds no expressive power without nonlinearities.

   (d) Backpropagation to get $\frac{\partial \mathcal{L}}{\partial x}$ typically requires much more time (e.g., many more additions and multiplications) than computing the original function $\mathcal{L}(x)$. True / False.

   > **Solution: False**: the backward pass costs a small constant factor (around two to three times) of the forward pass.

   (e) Backpropagation to get $\frac{\partial \mathcal{L}}{\partial x}$ typically requires much more memory (e.g., many more numbers stored at once) than computing the original function $\mathcal{L}(x)$. True / False.

   > **Solution: True**: the forward pass alone can discard each intermediate value once it is used, but backpropagation must store the activations of every layer for the backward pass.

   (f) 100% accuracy on the training set tells us the network generalizes well. True / False

   > **Solution: False**: the network may simply have memorized the training set.

   (g) 100% accuracy on the training set tells us the network does **not** generalize. True / False

   > **Solution: False**: a network can fit its training set perfectly and still generalize well.

   (h) 100% accuracy on a holdout test set tells us the network generalizes well. True / False

   > **Solution: True**: a holdout set drawn from the same distribution measures generalization (given a test set of reasonable size that was never used for training or tuning).

(i) 100% accuracy on a holdout test set tells us the network does **not** generalize. True / False

Solution: False: see the previous part.

(j) Enlarging the training set will tend to reduce the train/test accuracy gap. True / False

Solution: True: more data makes memorization harder and training accuracy closer to test accuracy.

(k) Enlarging each network layer will tend to reduce the train/test accuracy gap. True / False

Solution: False: more capacity tends to make overfitting easier, which widens the gap.

(l) Data augmentation will tend to reduce the train/test accuracy gap. True / False

Solution: True: it effectively enlarges the training set.

(m) Lower learning rates will tend to reduce the train/test accuracy gap. True / False

Solution: False: the learning rate is not a regularizer; if anything, the noise of large-step SGD has a mild regularizing effect.

(n) Dropout layers will tend to reduce the train/test accuracy gap. True / False

Solution: True: dropout is a regularizer.

(o) Xavier weight initialization will tend to reduce the train/test accuracy gap. True / False

Solution: False: initialization affects how well training proceeds, not overfitting.

(p) Xavier weight initialization will help avoid vanishing gradients. True / False

Solution: True: it scales the weights so that activation and gradient variances are preserved from layer to layer.

(q) Weight decay will help avoid vanishing gradients. True / False.

Solution: False: it shrinks weights, which, if anything, shrinks gradients through the layers.

(r) Batch normalization will help avoid vanishing gradients. True / False.

Solution: True: renormalizing the activations at each layer keeps the gradients well scaled.

(s) The sigmoid nonlinearity will help avoid vanishing gradients. True / False.

Solution: False: its derivative is at most 1/4 and nearly zero when saturated, which causes vanishing gradients.

(t) Enlarging the batch size will help avoid vanishing gradients. True / False.

Solution: False: a larger batch reduces gradient noise but does not change the gradient's magnitude through the layers.

(u) As long as the gradient is nonzero, updating parameters with a small enough learning rate guarantees a decrease in the loss. True / False.

Solution: True: for a smooth loss, a small enough step in the negative gradient direction decreases the loss to first order (for the full-batch loss).

(v) As long as the gradient is nonzero, updating parameters with large enough momentum guarantees a decrease in the loss. True / False.

Solution: False: large momentum can carry the parameters past the minimum and increase the loss.

(w) If you are using plain SGD, then dividing the loss by 10 before doing backpropagation will have the same effect as reducing learning rate by a factor of 10. True / False.

Solution: True: the gradient scales by 1/10, so the update $-\lambda \nabla \mathcal{L}$ is exactly the same as dividing λ by 10.

(x) If you are using momentum, dividing the loss by 10 will have little effect. True / False.

Solution: False: momentum is linear in the gradients, so this is again equivalent to dividing the learning rate by 10.

(y) If are using ADAM, dividing the loss by 10 will have little effect. True / False.

Solution: True: ADAM divides the first moment by the square root of the second moment, so the scale cancels (apart from the tiny ϵ).

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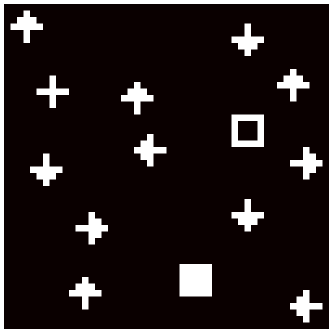
2. Convolutions

Pytorch defines a 1-channel 5x5 convolution with padding of 2 as:

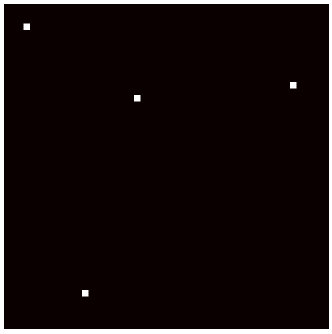
$$\text{output}_{x,y} = \text{bias} + \sum_{i=0}^{4}\sum_{j=0}^{4} \text{weight}_{i,j} \cdot \text{input}_{x+i-2,y+j-2} \tag{1}$$

Your goal is to create a small neural network using this kind of convolution that transforms the grid of input data on the left to the output data in the middle. The input grid was formed by placing a bunch of 5x5 emoji-style symbols as +1 patterns in a field of −1 values, and every point in both the input and output data is ±1. An overlay of the input and desired output are shown on the right.

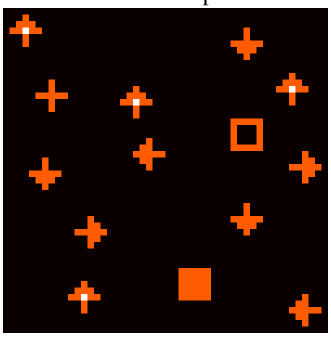
Given input:
1x1x50x50 tensor
white = +1, black = -1



Desired output:
1x1x50x50 tensor
white = +1, black = -1



Detail: overlay of
given input and
desired output



(a) The following code solves the problem, but the specific numbers for the convolution weights and bias need to be supplied. Fill in numbers that will make the program produce the desired output. Hint: you can use ±1 for most of the numbers.

```
import torch
torch.set_grad_enabled(False)

conv = torch.nn.Conv2d(
    in_channels  = 1,
    out_channels = 1,
    kernel_size  = (5,5),
    padding      = 2)

class Sign(torch.nn.Module):
    def forward(self, x):
        return x.sign()

net = torch.nn.Sequential(
    conv,
    Sign()
)

input_data = torch.load('input_data.pt')
```

```
conv.weight[:,:,:,:] = torch.tensor([[[
    [ ,  ,  ,  ,  ],
    [ ,  ,  ,  ,  ],
    [ ,  ,  ,  ,  ],
    [ ,  ,  ,  ,  ],
    [ ,  ,  ,  ,  ],
]]])
conv.bias[:] = torch.tensor(
    [ ]
)
output = net(input_data)
```

Solution: The output marks the center of every copy of one particular symbol, the cross with a wider top (the "arrow" shape whose first two rows are a dot and a bar of three). Since the formula indexes $\text{input}_{x+i-2,\,y+j-2}$ with no flip, the weights are a template of that symbol: +1 where the symbol is white and −1 where it is black. Each row of the tensor literal is one row of the image (PyTorch stores the vertical axis

first), so the matrix looks like the symbol, upright.

$$\text{weight} = \begin{bmatrix} -1 & -1 & 1 & -1 & -1 \\ -1 & 1 & 1 & 1 & -1 \\ 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 & -1 \end{bmatrix}, \qquad \text{bias} = -23.$$

A perfect match scores 25. Every mismatched pixel lowers the score by 2, and the closest other symbols (the plain cross and the cross with a wider bottom) differ in two pixels, scoring at most 21. Any bias strictly between -25 and -21 works, such as -23 or -24; then sign gives $+1$ only at the centers of the target symbol. (Checked numerically against the input grid.)

(b) This particular convolution has some symmetry in the way it acts on its data. Circle which types of symmetry it has.

Invariant under horizontal reflection $(x, y) \leftrightarrow (-x, y)$	Equivariant under horizontal reflection $(x, y) \leftrightarrow (-x, y)$
Invariant under 180 degree rotation	Equivariant under 180 degree rotation
Invariant under translation	Equivariant under translation

Solution: Equivariant under horizontal reflection and **equivariant under translation**. Every convolution is translation-equivariant (away from the padded border): shifting the input shifts the output. This kernel is left-right symmetric, so mirroring the input mirrors the output. It is not symmetric under 180-degree rotation, which turns the target into the wider-bottom cross that the filter rejects. None of the invariances hold: moving or mirroring a symbol moves its output dot.

3. Weight decay

Weight decay adds a term to the loss that sums the square of all the model parameters w_i:

$$\mathcal{L}_{\text{wd}} = \delta \sum_i ||w_i||^2 \tag{2}$$

(a) What is the purpose of weight decay?

Solution: Regularization. It penalizes large weights, pulling the model toward small-weight, simpler and smoother functions, which reduces overfitting and narrows the train/test gap.

(b) Suppose that weight decay is the only term in the loss, $\mathcal{L} = \mathcal{L}_{\text{wd}}$, and we have set weight decay $\delta = 0.1$ and learning rate $\lambda = 0.1$, using ordinary gradient descent that updates parameters in the direction of the gradient:

$$w_i^{(\text{updated})} := w_i - \lambda \frac{\partial \mathcal{L}}{\partial w_i} \tag{3}$$

What is the value of $w_i^{(\text{updated})}$ after one iteration of gradient descent? Write a simple expression in terms of the original w_i.

Solution: $\partial \mathcal{L}/\partial w_i = 2\delta w_i = 0.2\, w_i$, so $w_i^{(\text{updated})} = w_i - 0.1 \cdot 0.2\, w_i = \boxed{0.98\, w_i}$.

(c) Now also suppose the network $z = f(x)$ is an ordinary feedforward network that outputs a vector of logits z, with several linear layers and the ReLU nonlinearity between each layer. What does $f(x)$ converge towards as the number of training iterations increases towards infinity if the whole loss is $\mathcal{L} = \mathcal{L}_{\text{wd}}$?

Solution: Each step multiplies every parameter by 0.98, so all weights and biases decay to zero and $f(x) \to \boxed{0}$, the zero vector of logits, for every input (a uniform softmax).

(d) Now suppose instead that $f(x)$ is a residual network with no bottleneck layers. In that case, what does $f(x)$ converge towards as training iterations go to infinity? Again assume the whole loss is $\mathcal{L} = \mathcal{L}_{\text{wd}}$.

Solution: Each residual block computes $x + g(x)$. As its weights go to zero, $g(x) \to 0$, so each block becomes the identity and $f(x) \to \boxed{x}$, the identity function.

4. Back propagation

The following computation graph computes the squared error of $\mathcal{L} = ||Wx - y||^2$. Your goal is to compute $\frac{\partial \mathcal{L}}{\partial W}$ as well as $\frac{\partial \mathcal{L}}{\partial x}$ using the backpropagation algorithm, for the specific values of W and x shown in the figure.

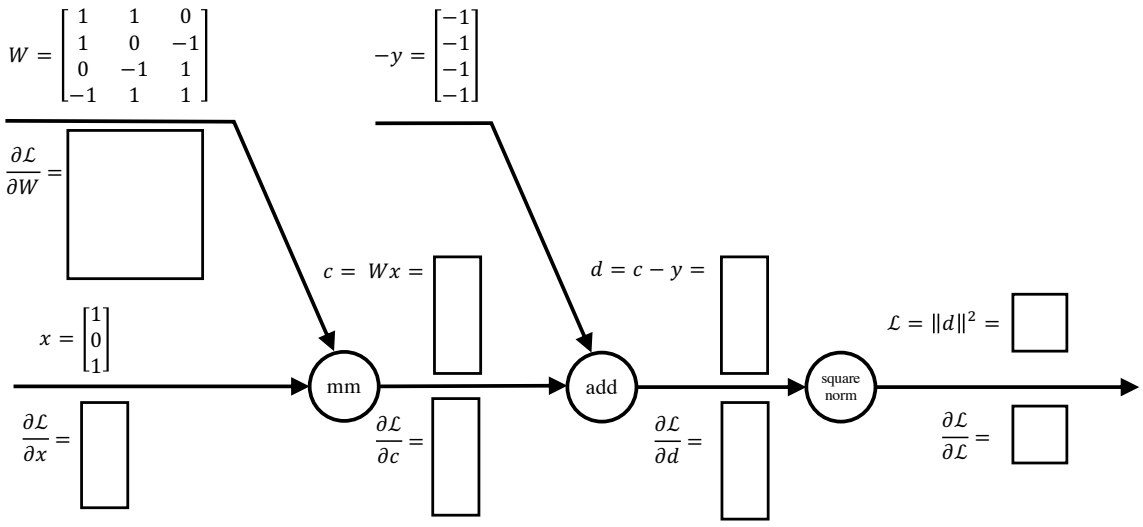
Keep in mind that $||d||^2$ is defined as

$$||d||^2 = \sum_i d_i^2 \tag{4}$$

And also keep in mind that the shape of the gradient with respect to any given tensor result is always the same as the shape of the forward result.

(a) Fill in the forward-pass in the figure below by filling in the values of the tensors above each arrow where there is a box.

Solution: With $x = [1, 0, 1]^\top$, multiplying by W gives $c = Wx = [1, 0, 1, 0]^\top$. Since $-y = [-1, -1, -1, -1]^\top$, the add gives $d = c - y = [0, -1, 0, -1]^\top$, and $\mathcal{L} = \|d\|^2 = 2$.



(b) Fill in the backward-pass in the figure above by filling in the boxes below the arrows.

Solution: $\dfrac{\partial \mathcal{L}}{\partial \mathcal{L}} = 1$, $\dfrac{\partial \mathcal{L}}{\partial d} = 2d = [0, -2, 0, -2]^\top$, $\dfrac{\partial \mathcal{L}}{\partial c} = \dfrac{\partial \mathcal{L}}{\partial d} = [0, -2, 0, -2]^\top$ (the add passes the gradient through),

$$\frac{\partial \mathcal{L}}{\partial W} = \frac{\partial \mathcal{L}}{\partial c} x^\top = \begin{bmatrix} 0 & 0 & 0 \\ -2 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & -2 \end{bmatrix}, \qquad \frac{\partial \mathcal{L}}{\partial x} = W^\top \frac{\partial \mathcal{L}}{\partial c} = -2 \left(\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ -2 \\ 0 \end{bmatrix}.$$

Shapes match the forward values: 4×3 for W, 3 for x, and 4 for c and d.

5. Pytorch modules

Read and think about the following code.

```
1  import torch
2  from torch.nn import Sequential, ReLU, Linear
3  # commented out: # torch.set_grad_enabled(False)
4  num_pixels = 1000000
5  bottleneck = 100
6  net = Sequential(
7      Linear(in_features=num_pixels, out_features=bottleneck, bias=True),
8      ReLU(),
9      Linear(in_features=bottleneck, out_features=num_pixels, bias=False)
10 ).cuda()
11 print('Parameter tensors', len(list(net.named_parameters())))
12 print('Parameter elements', sum(p.numel() for p in net.parameters()))
13 total_error = 0
14 sample_size = 10000
15 for test_index in range(sample_size):
16     test_data = torch.randn(1, num_pixels, device='cuda')
17     total_error += (net(test_data) - test_data).pow(2).mean()
18 print('Average error', total_error / sample_size)
```

(a) Is this network being used as a classifier or an autoencoder? How can you tell?

Solution: An **autoencoder**. The output has the same size as the input (10^6), there is a narrow 100-unit bottleneck in the middle, and the error on line 17 compares `net(test_data)` with `test_data` itself rather than with any labels.

(b) Think about which parameters are contained in a Linear layer, either when it has a bias term, or when it does not. What is the count of parameter tensors that you expect to be printed on line 11?

Solution: 3: the first Linear layer's weight and bias, and the second Linear layer's weight (it has `bias=False`). ReLU has no parameters.

(c) Think about how big the parameter matrices are in a Linear layer. What is the total count of parameters that you expect to be printed on line 12?

Solution: 100×10^6 (weight 1) + 100 (bias 1) + $10^6 \times 100$ (weight 2) = $\boxed{200{,}000{,}100}$.

(d) When you run this code, it prints out values on lines 12 and 13 quickly but then it always crashes after about 10 seconds, saying 'CUDA out of memory' before getting to line 18. Why is it running out of memory? How could you fix the code so that it easily runs 10,000 iterations using the GPU without changing the network or the data; it should be able to quickly print out something like 'Average error 1.055' on line 18?

Solution: Gradients are enabled (line 3 is commented out), so each `net(test_data)` builds an autograd graph that keeps its intermediate activations (several million-element tensors) for a possible backward pass. Line 17 adds each loss *tensor* into `total_error`, which keeps every iteration's graph alive, so the memory grows by several megabytes per iteration until the GPU runs out. **Fix:** disable gradients, e.g. uncomment `torch.set_grad_enabled(False)` or wrap the loop in `with torch.no_grad():`, or accumulate a plain number with `total_error += (...).mean().item()`, which drops each graph after its iteration.

6. Softmax and cross-entropy loss

Label smoothing is a classification objective that reduces the overconfidence of a model by moderating the ordinary classification goal. Here is how it works: when given training data x labeled with class c, the goal is not to give that class a probability of one, but instead to aim for probability $y_c = 1 - \alpha$, while distributing the rest of the probability to other classes evenly.

For example, if $\alpha = 0.1$ and we have two classes, then the target probability of something in class 2 will be $y_2 = 0.9$ and $y_1 = 0.1$. When using label smoothing, we still use the typical softmax cross-entropy loss, so that we wish to learn a model $z = f(x)$ that computes logits z from data x, so that the predictions $p = \text{softmax}(z)$ minimize the loss $\mathcal{L}_{\text{ce}} = \text{CE}(y; p)$.

In other words, these ordinary definitions still apply:

$$\mathcal{L}_{\text{ce}} = \text{CE}(y; p) = -\sum_i y_i \log p_i \tag{5}$$

$$p_j = \text{softmax}(z)_j = \frac{e^{z_j}}{\sum_i e^{z_i}} \tag{6}$$

Now, set label smoothing with $\alpha = 0.1$ for a classification problem with $n = 2$ classes. Our network will output two logits $[z_1, z_2] = f(x)$ for any input x. Each part of the question below will examine a different data point x, all which have the true class label $c = 2$.

(a) Suppose x is in class $c = 2$. Then think about the target y_1 and y_2 defined above, and which probability vector $p = \text{softmax}(f(x))$ would minimize $\mathcal{L}_{\text{ce}} = \text{CE}(y; p)$. What value should the predicted probability p_2 take to minimize the cross-entropy loss?

Solution: Cross-entropy $\text{CE}(y; p)$ is minimized when $p = y$, so $p_2 = y_2 = \boxed{0.9}$ (and $p_1 = 0.1$).

(b) Now consider another point x that also has class $c = 2$, and suppose the first logit is $z_1 = 0$ (i.e., the number zero). In part (a) you figured out what p_2 must be. Now that you know p_2, what value must the second logit z_2 have? Don't use a calculator: you can express your answer in terms of log or exp.

Solution: $p_2 = \dfrac{e^{z_2}}{e^0 + e^{z_2}} = 0.9$ gives $e^{z_2} = 9$, so $z_2 = \boxed{\log 9} \approx 2.2$. (Equivalently, $z_2 - z_1 = \log(p_2/p_1) = \log 9$.)